

Benchmarking Report

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1. Benchmarking report

2. Algorithms

The following algorithms are considered in this benchmarking report.

2.1 L-BFGS-B

Limited-memory BFGS algorithm implemented in the SciPy library

2.2 SLSQP

Sequential Least-Squares Quadratic Programming (SLSQP) implemented in the SciPy library

3. Problem configurations

The benchmarking reference problems are listed below.

- [Rastrigin](#)
- [Rosenbrock](#)
- [Rosenbrock 5D](#)

4. Results

The results of each group of algorithms configurations on each group of problem configurations.

- [L-BFGS-B_configurations](#)
- [SLSQP](#)

5. Problems

5.1 Rastrigin

5.1.1 Description

No description available.

Optimal feasible objective value: 0.

5.1.2 Target values

- 0.0001 (feasible)
- 1e-05 (feasible)
- 1e-06 (feasible)
- 0.0 (feasible)

5.2 Rosenbrock 5D

5.2.1 Description

No description available.

Optimal feasible objective value: 0.

5.2.2 Target values

- 0.0001 (feasible)
- 1e-05 (feasible)
- 1e-06 (feasible)
- 0.0 (feasible)

5.3 Rosenbrock

5.3.1 Description

No description available.

Optimal feasible objective value: 0.

5.3.2 Target values

- 0.0001 (feasible)
- 1e-05 (feasible)
- 1e-06 (feasible)
- 0.0 (feasible)

6. Results

6.1 L-BFGS-B configurations

- [2D_problems](#)

6.2 SLSQP

- [2D_problems](#)

6.3 L BFGS B configurations

6.3.1 2D problems

The results of the group of algorithms configurations "L-BFGS-B configurations" on the group of problems "2D problems".

The algorithms configurations

- L-BFGS-B with 2 Hessian corrections
- L-BFGS-B with 20 Hessian corrections

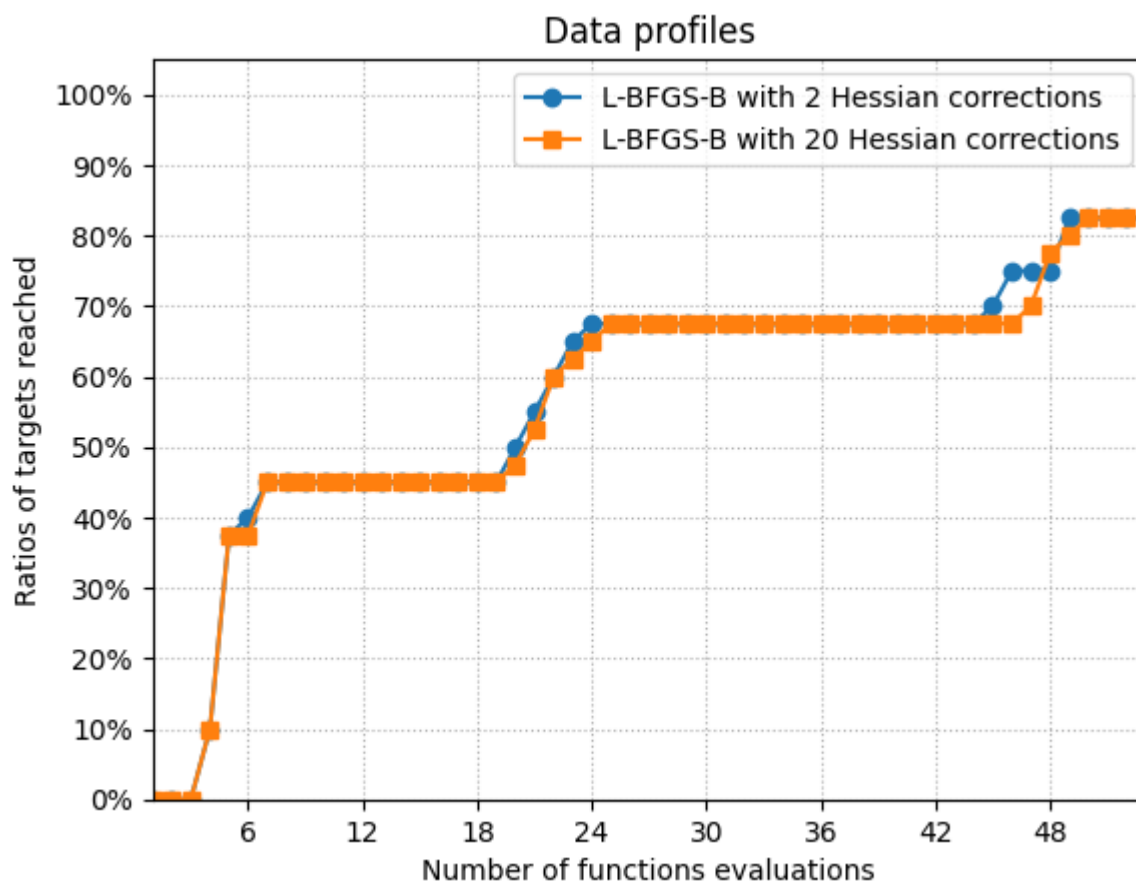
The problems

- Rastrigin
- Rosenbrock

Benchmarking results

GLOBAL RESULTS

The performances of the algorithms on the reference problems of the group "2D problems" are represented in the following data profile.



RESULTS FOR EACH PROBLEM

The results of the algorithms configurations for each problem are linked below.

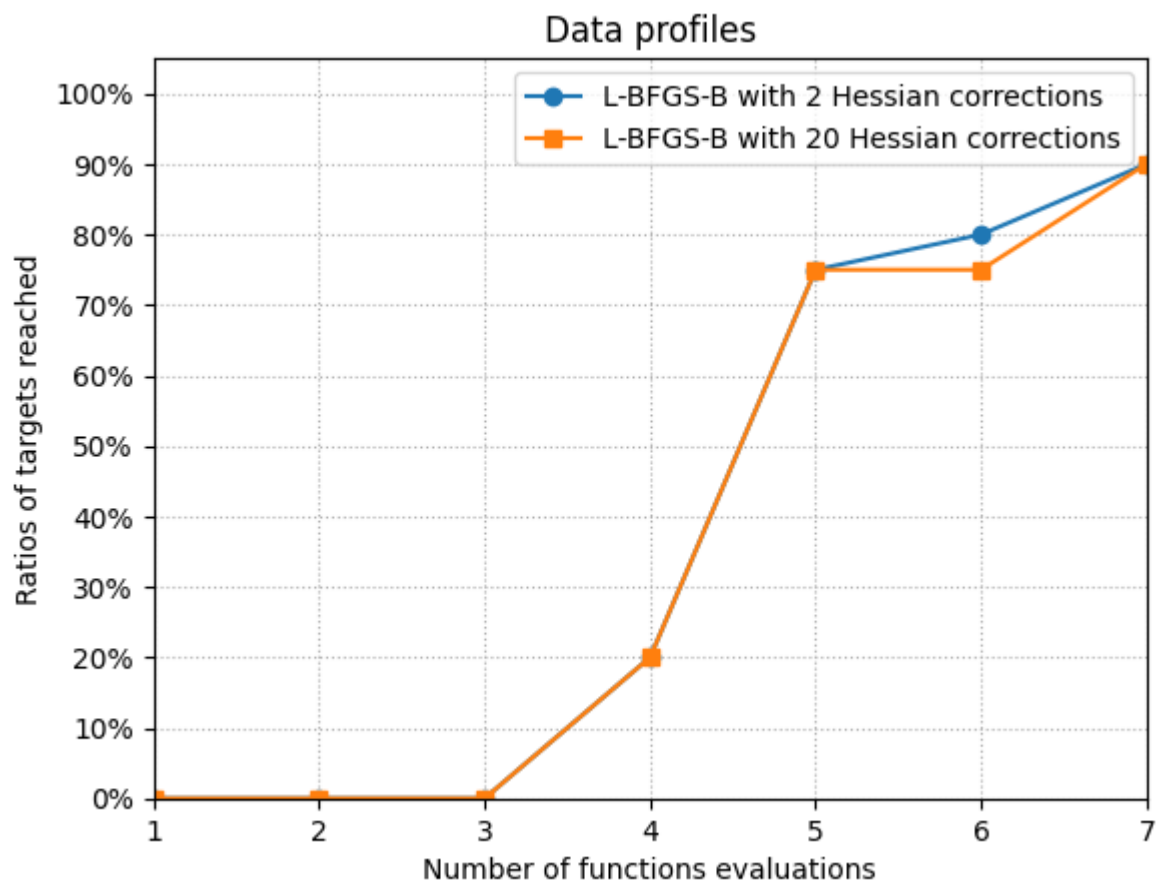
- [Rastrigin](#)
- [Rosenbrock](#)

6.3.2 2D problems

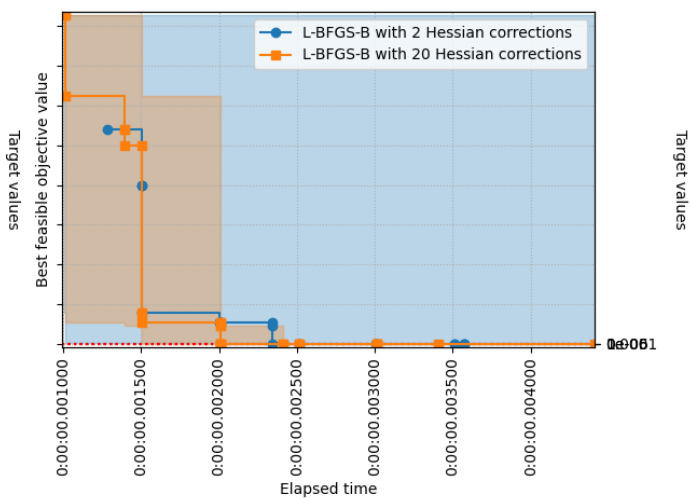
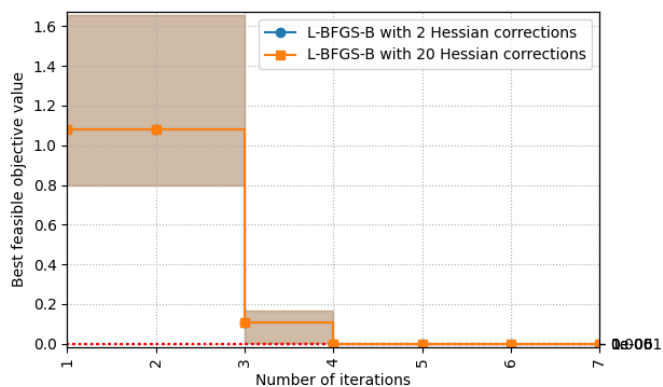
Rastrigin

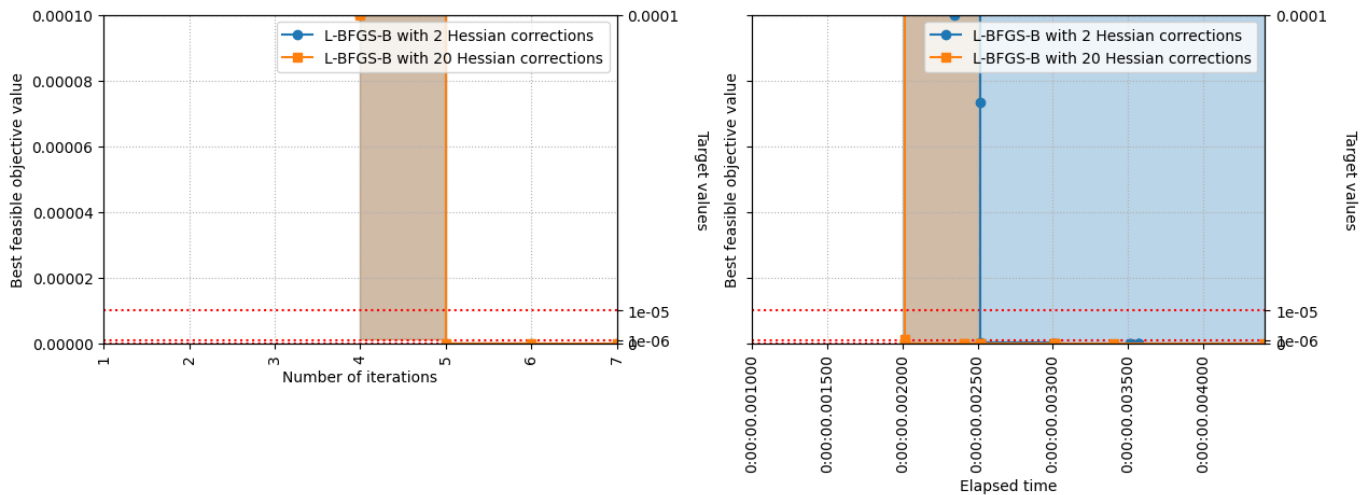
The results of the group of algorithms configurations 'L-BFGS-B configurations' for problem configuration 'Rastrigin'.

DATA PROFILES



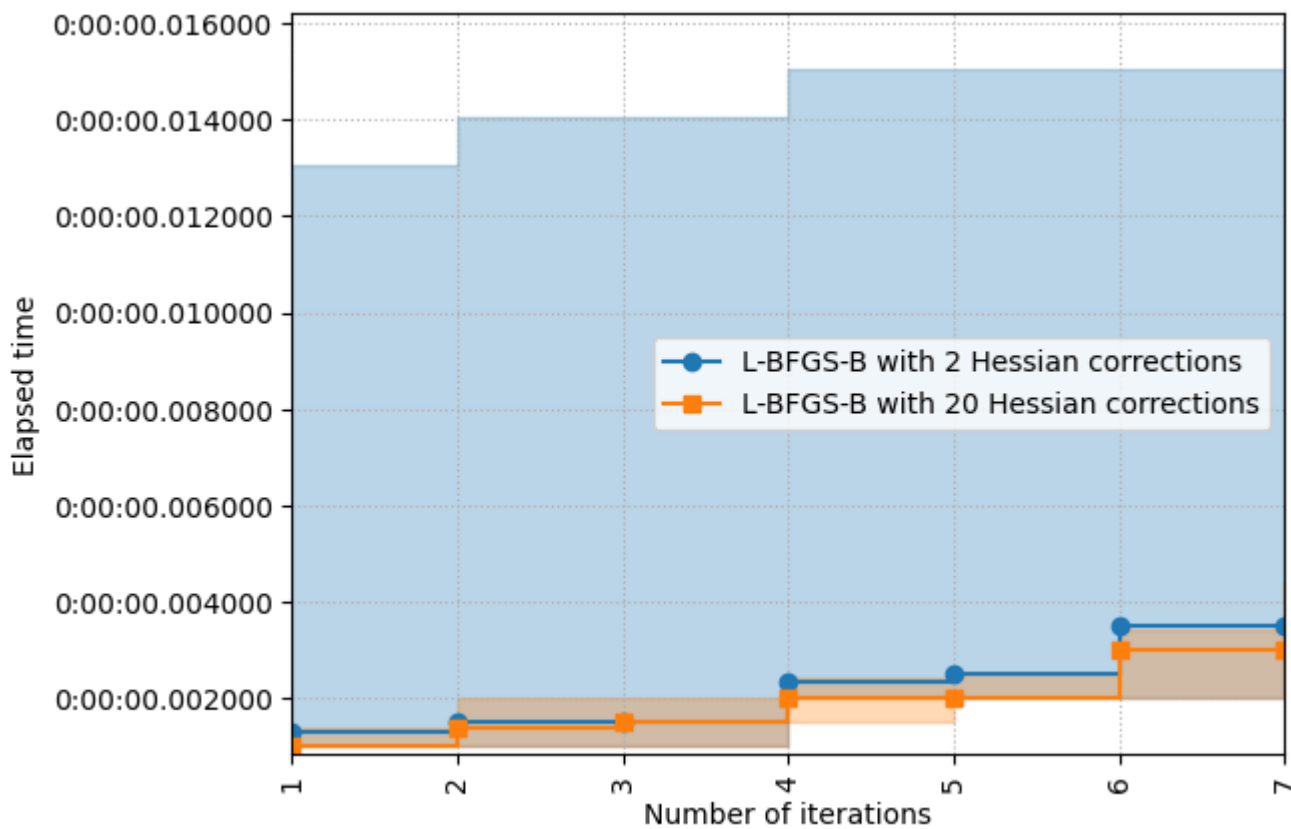
BEST FEASIBLE OBJECTIVE VALUE





	minimum	25th centile	median	75th centile	maximum
L-BFGS-B with 2 Hessian corrections	0	0	0	3.55271e-15	5.32907e-15
L-BFGS-B with 20 Hessian corrections	0	0	0	1.77636e-15	5.32907e-15

EXECUTION TIME



RESULTS FOR EACH ALGORITHM CONFIGURATION

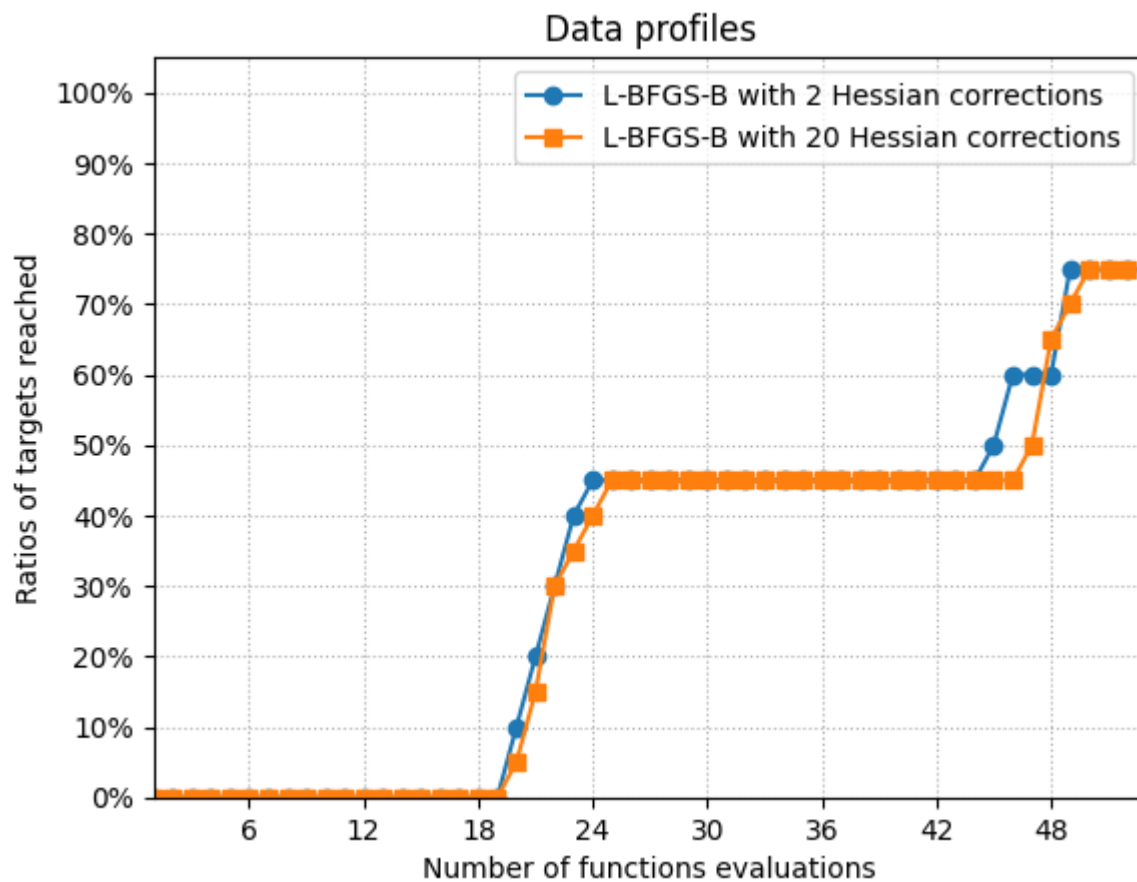
- [L-BFGS-B_with_2_Hessian_corrections](#)

- L-BFGS-B_with_20_Hessian_corrections

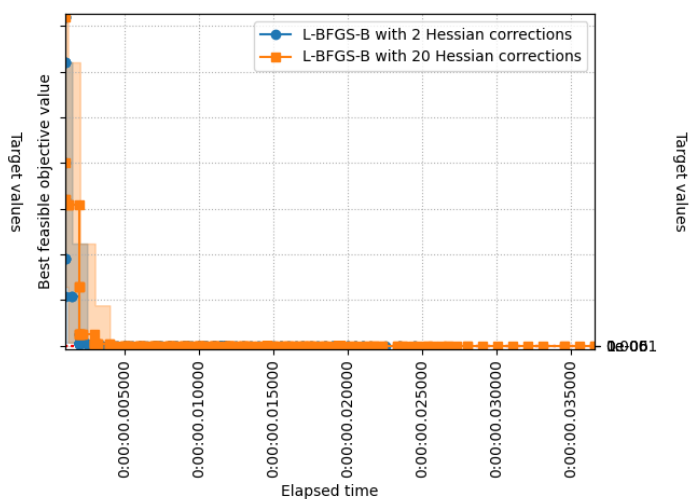
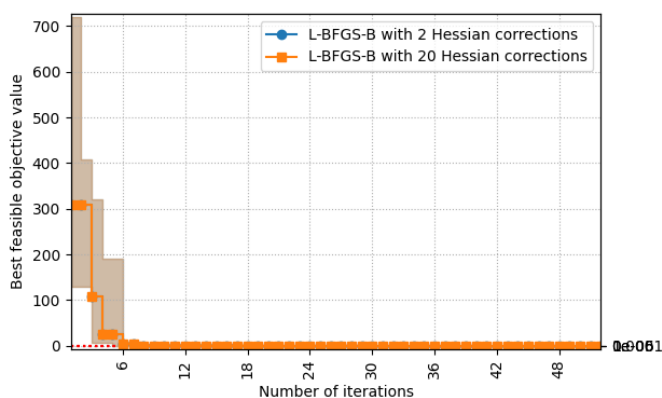
Rosenbrock

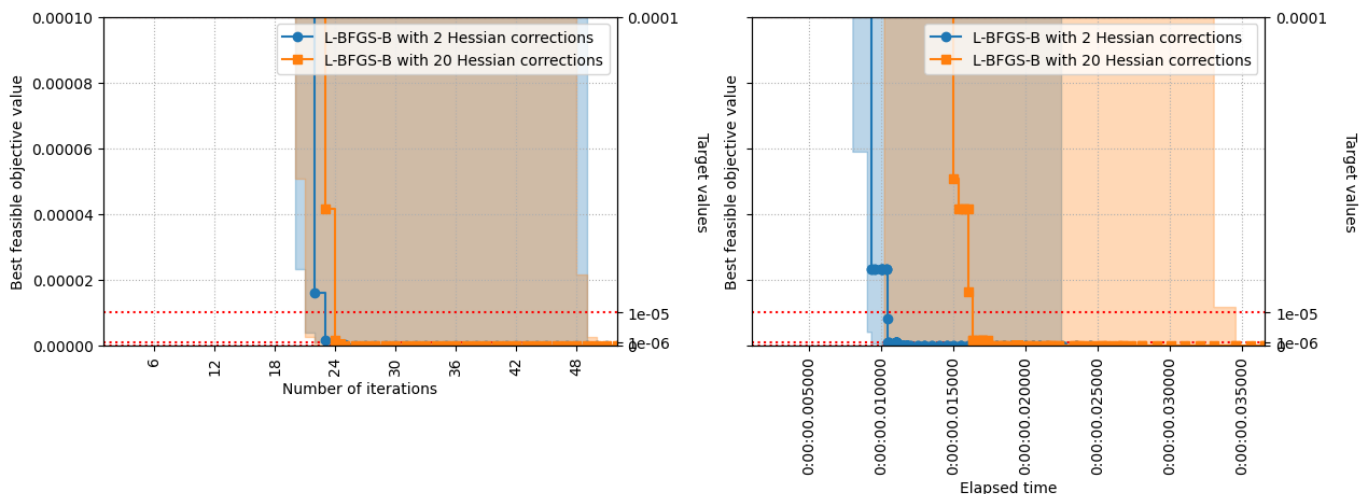
The results of the group of algorithms configurations 'L-BFGS-B configurations' for problem configuration 'Rosenbrock'.

DATA PROFILES



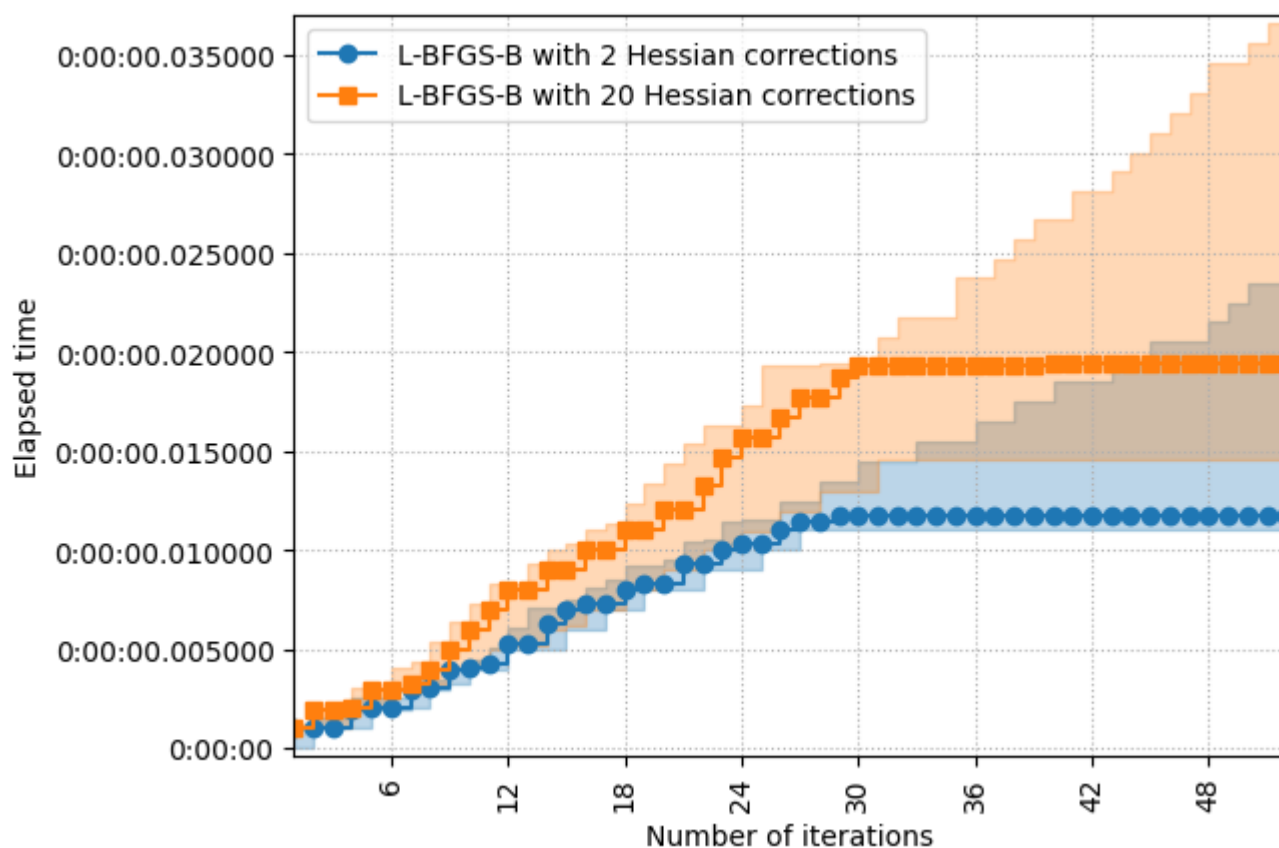
BEST FEASIBLE OBJECTIVE VALUE





	minimum	25th centile	median	75th centile	maximum
L-BFGS-B with 2 Hessian corrections	1.74156e-20	1.19054e-17	1.13779e-15	3.3576e-15	4.68664e-13
L-BFGS-B with 20 Hessian corrections	3.39836e-20	2.33591e-19	1.4195e-18	1.76694e-18	6.79141e-14

EXECUTION TIME



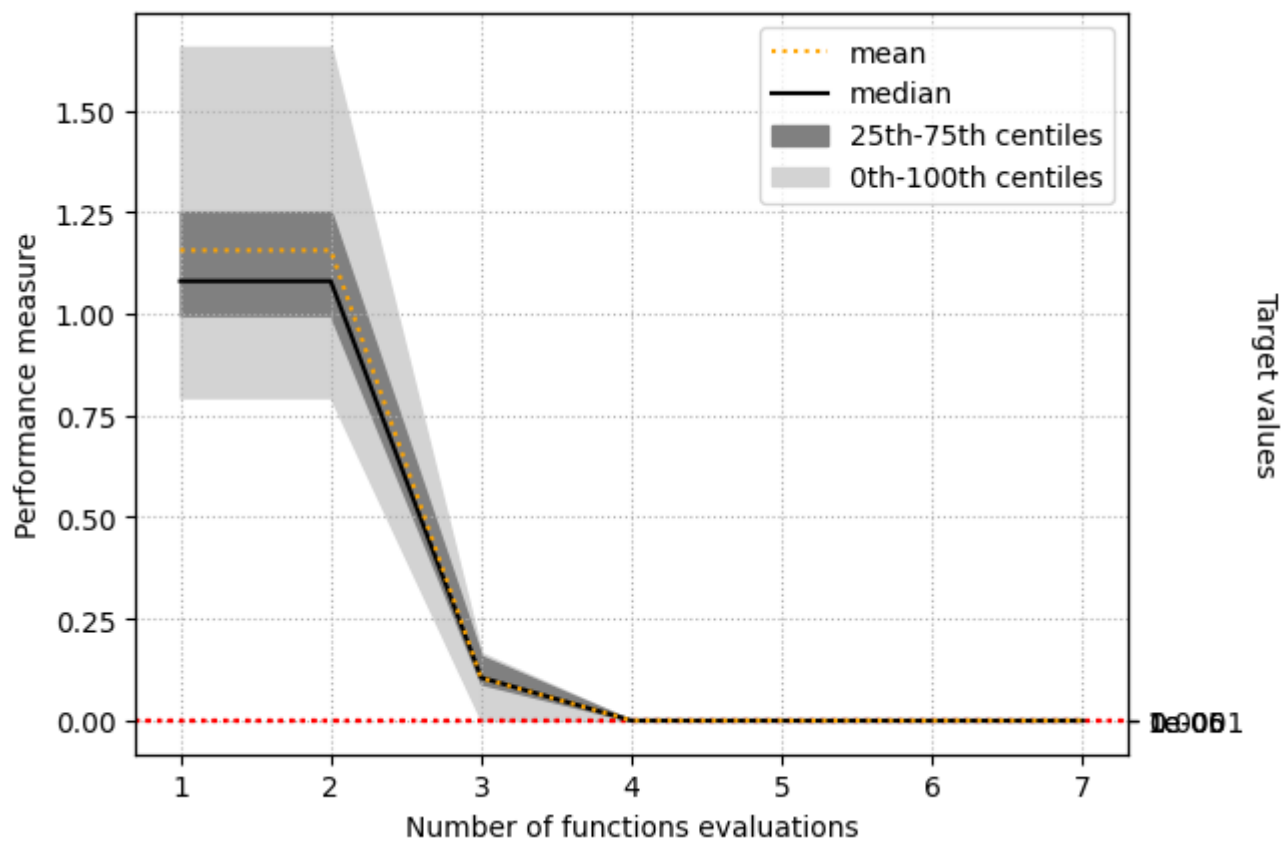
RESULTS FOR EACH ALGORITHM CONFIGURATION

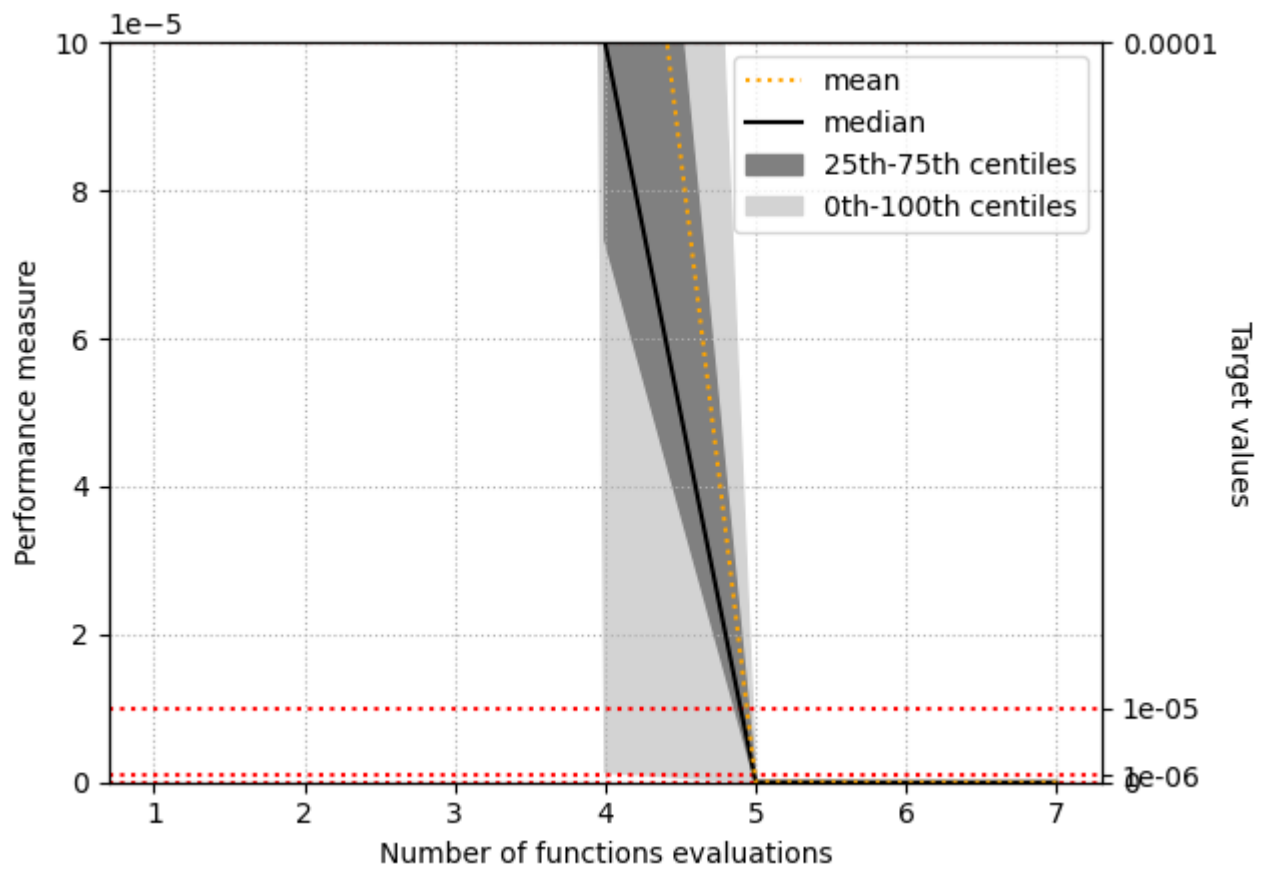
- L-BFGS-B_with_2_Hessian_corrections
- L-BFGS-B_with_20_Hessian_corrections

Rastrigin

L-BFGS-B WITH 20 HESSIAN CORRECTIONS ON RASTRIGIN

Performance measure

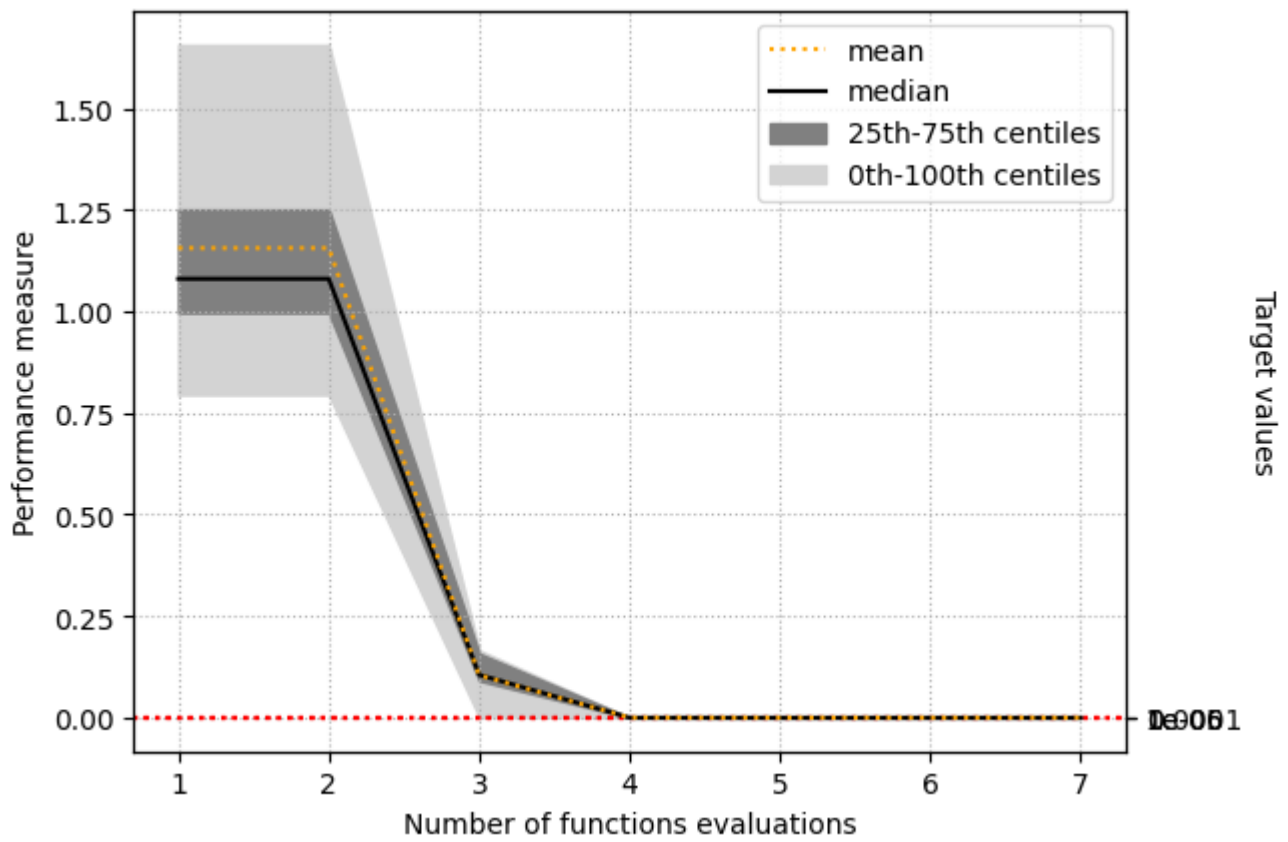


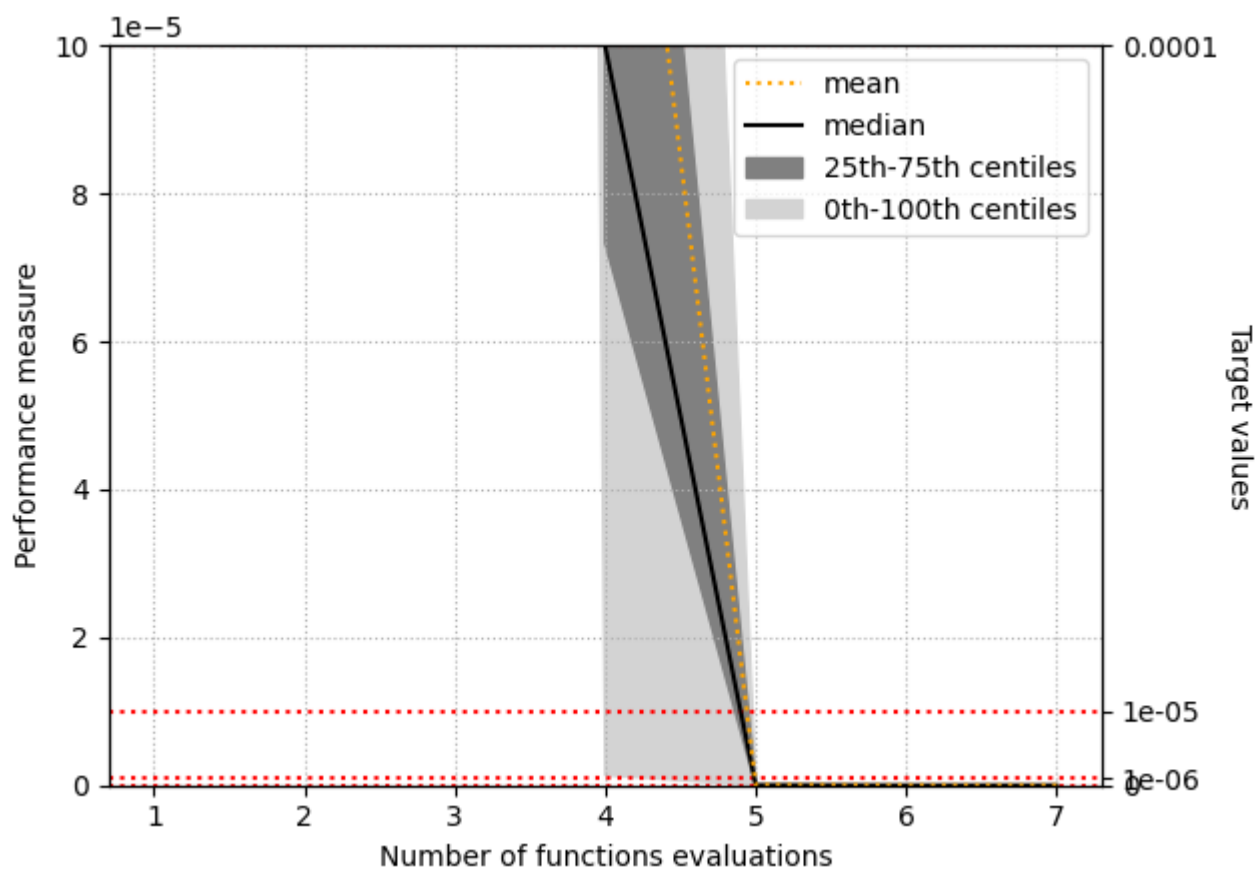


L-BFGS-B with 20 Hessian corrections	
maximum	5.32907e-15
75th centile	1.77636e-15
median	0
25th centile	0
minimum	0

L-BFGS-B WITH 2 HESSIAN CORRECTIONS ON RASTRIGIN

Performance measure



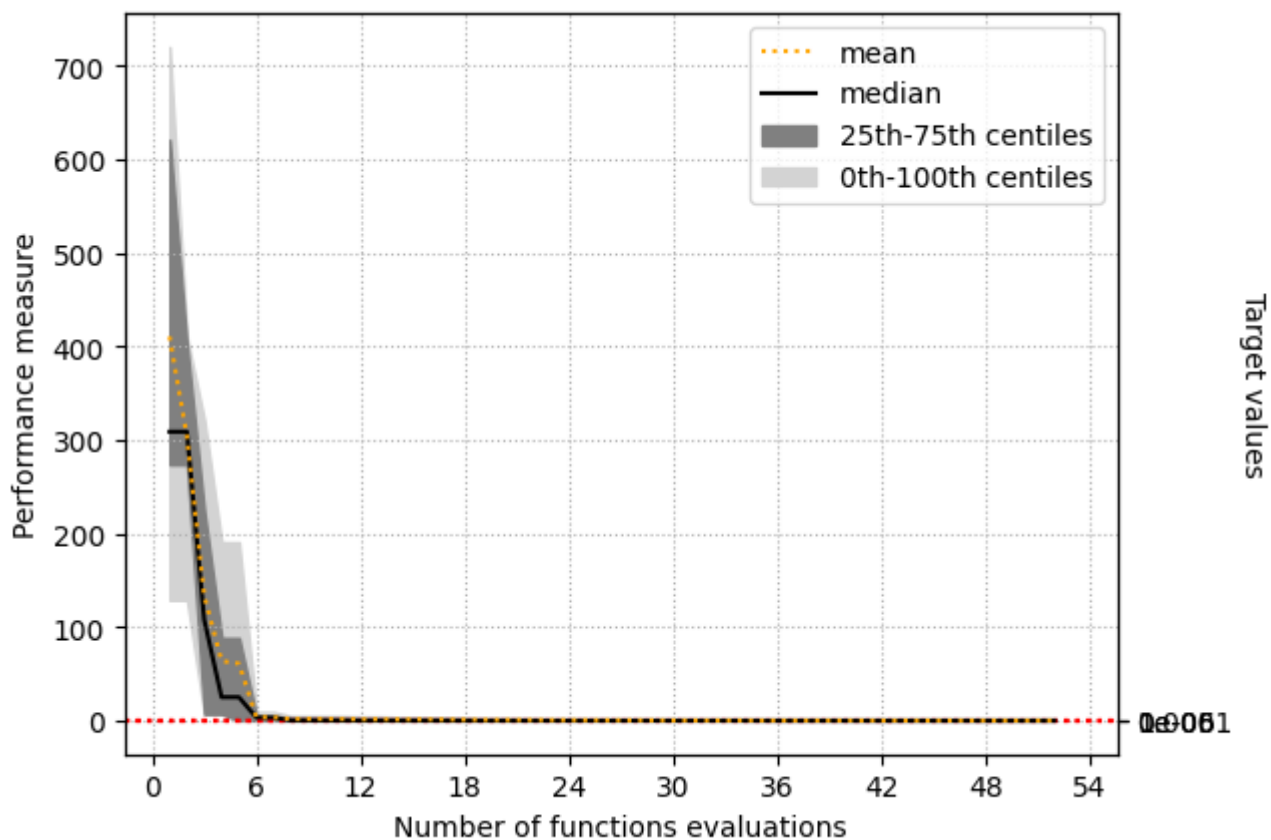


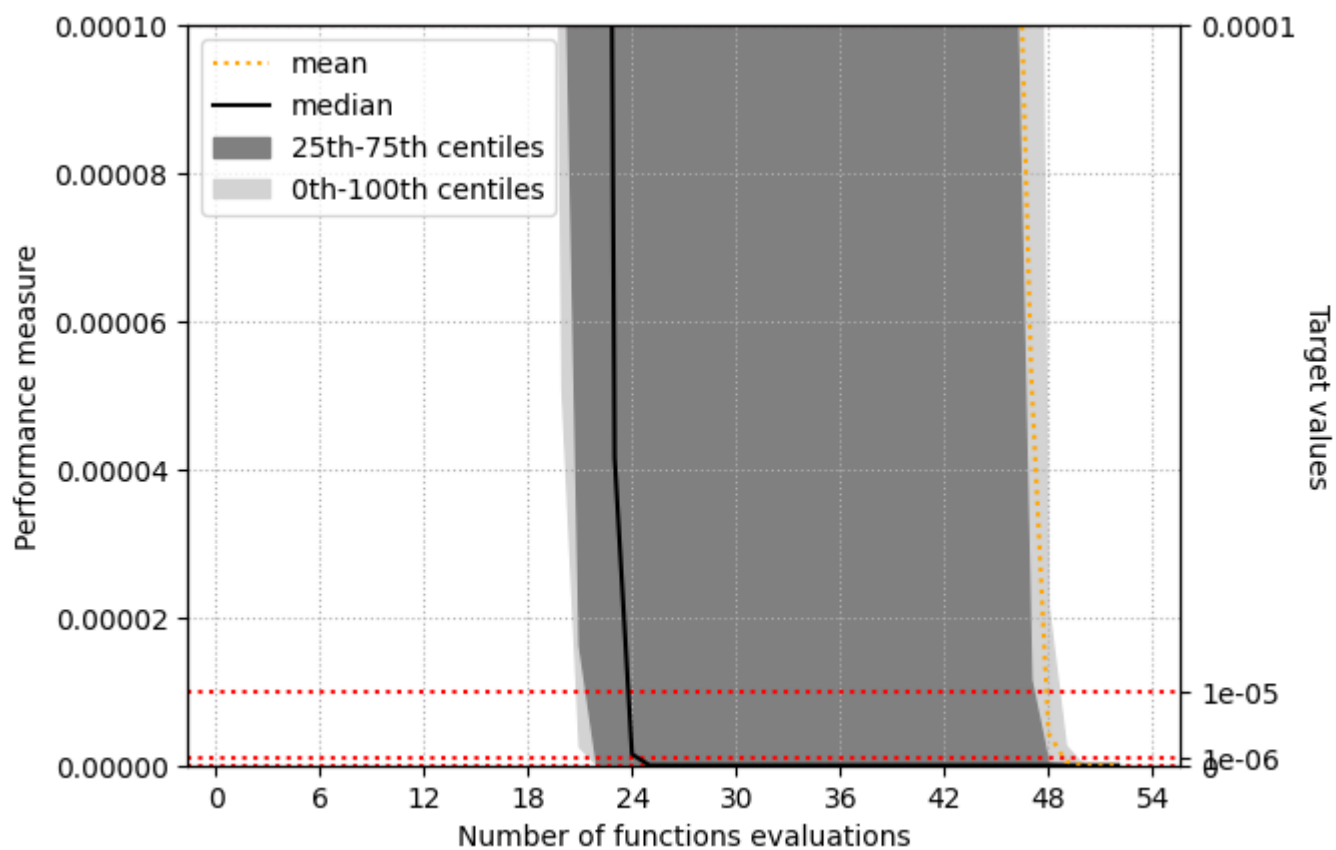
L-BFGS-B with 2 Hessian corrections	
maximum	5.32907e-15
75th centile	3.55271e-15
median	0
25th centile	0
minimum	0

Rosenbrock

L-BFGS-B WITH 20 HESSIAN CORRECTIONS ON ROSEN BROCK

Performance measure

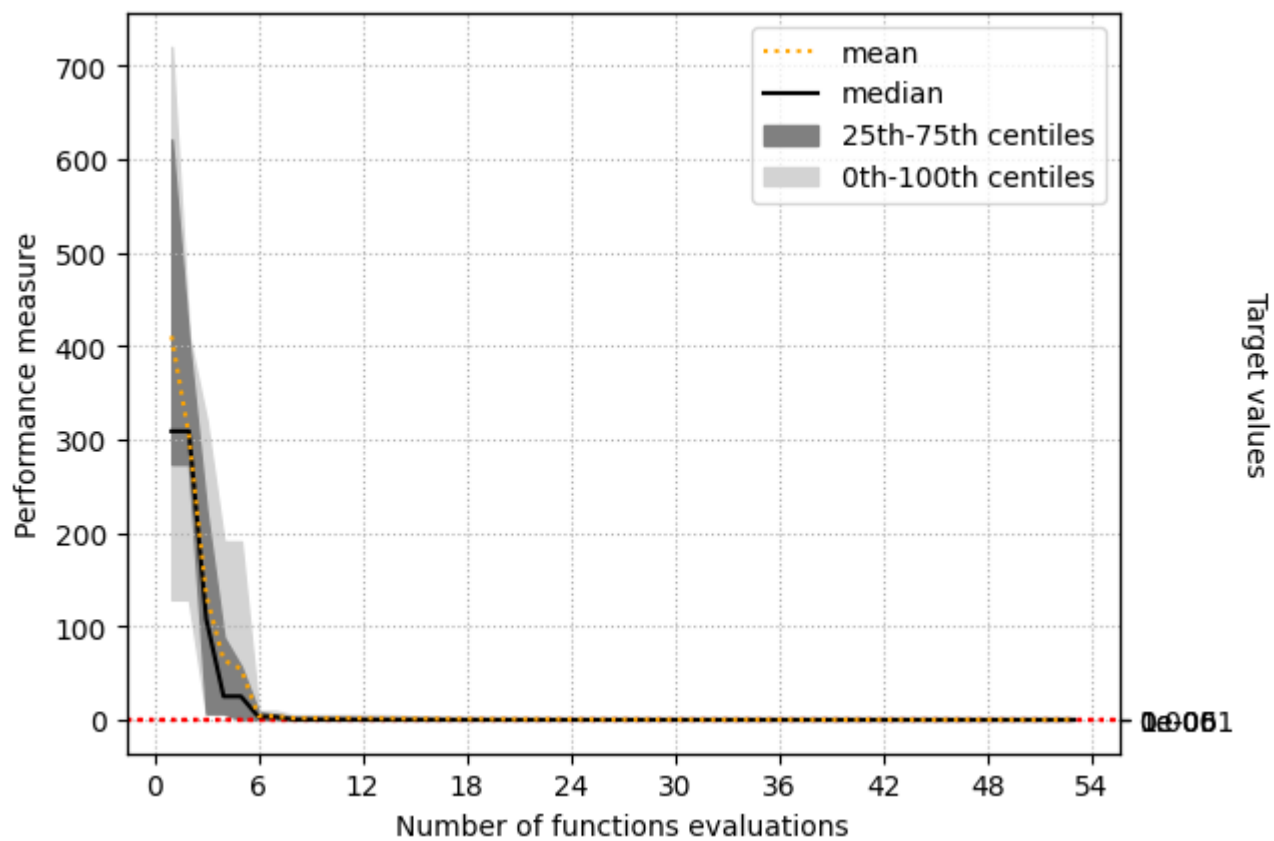


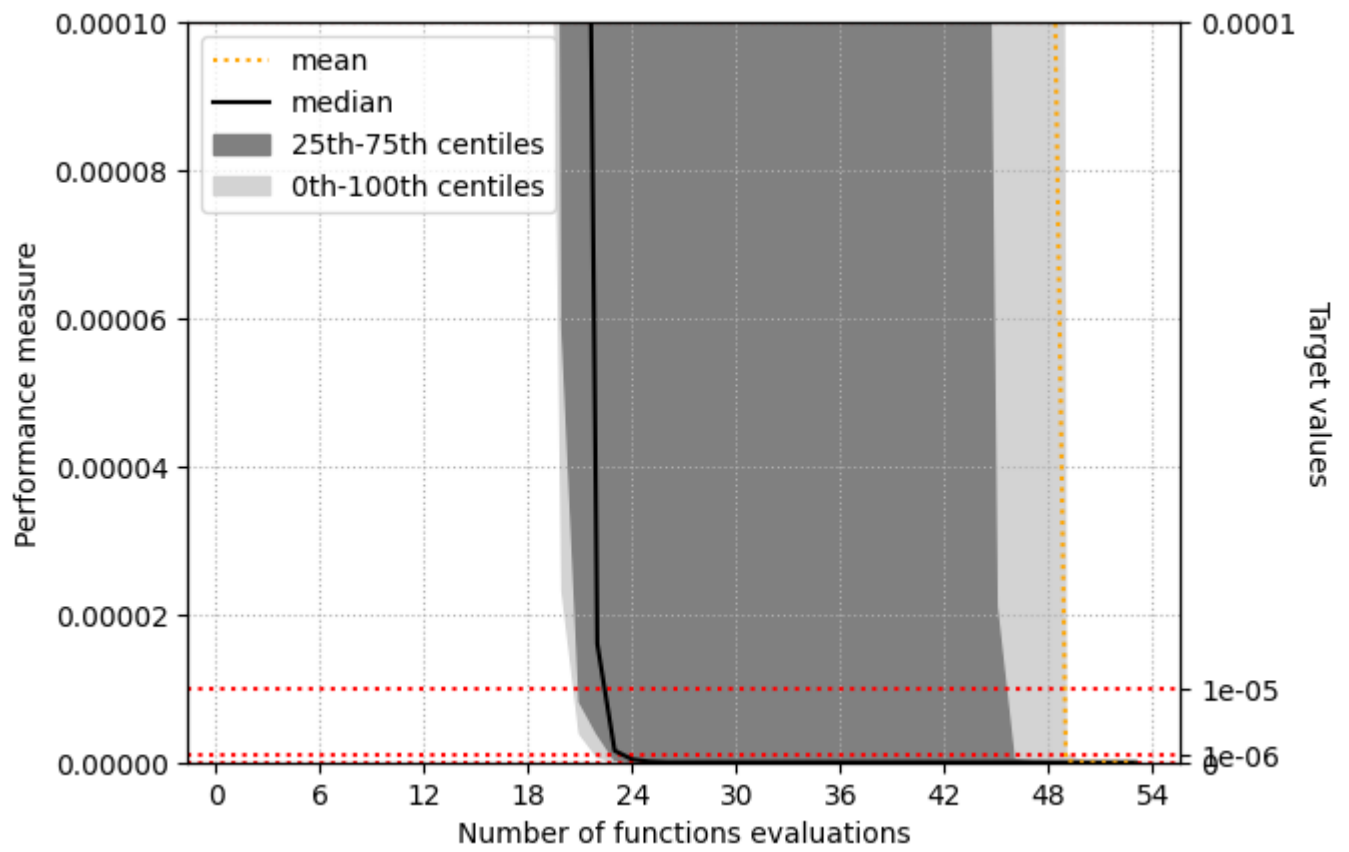


L-BFGS-B with 20 Hessian corrections	
maximum	6.79141e-14
75th centile	1.76694e-18
median	1.4195e-18
25th centile	2.33591e-19
minimum	3.39836e-20

L-BFGS-B WITH 2 HESSIAN CORRECTIONS ON ROSENBROCK

Performance measure





L-BFGS-B with 2 Hessian corrections	
maximum	4.68664e-13
75th centile	3.3576e-15
median	1.13779e-15
25th centile	1.19054e-17
minimum	1.74156e-20

6.4 SLSQP

6.4.1 2D problems

The results of the group of algorithms configurations "SLSQP" on the group of problems "2D problems".

The algorithms configurations

- SLSQP

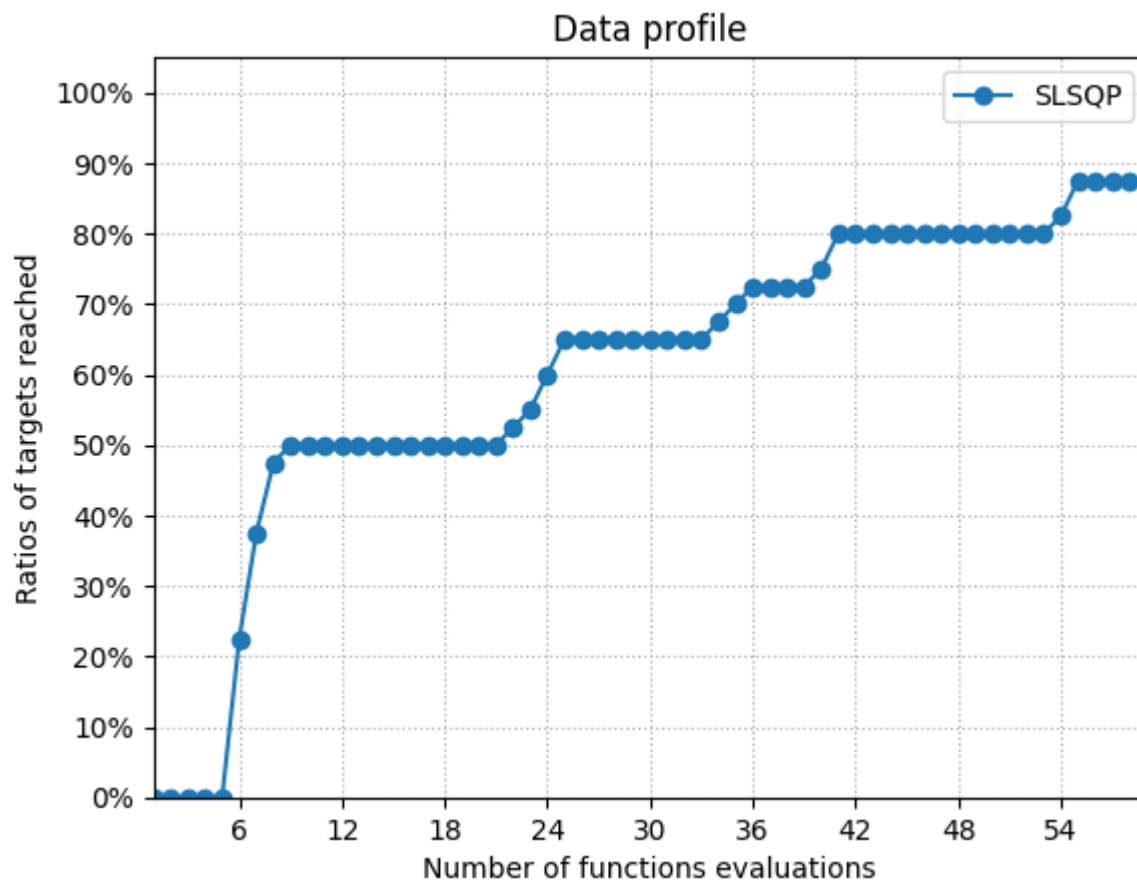
The problems

- Rastrigin
- Rosenbrock

Benchmarking results

GLOBAL RESULTS

The performances of the algorithms on the reference problems of the group "2D problems" are represented in the following data profile.



RESULTS FOR EACH PROBLEM

The results of the algorithms configurations for each problem are linked below.

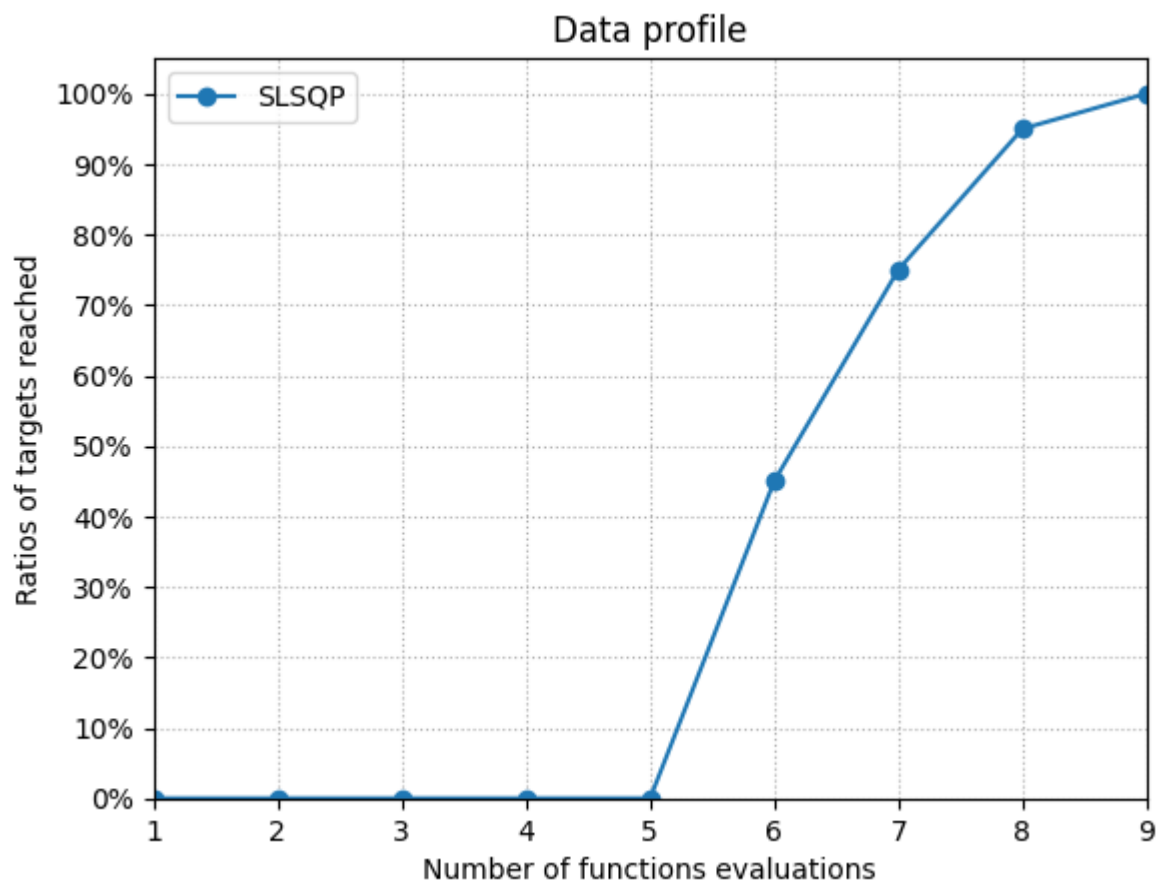
- [Rastrigin](#)
- [Rosenbrock](#)

6.4.2 2D problems

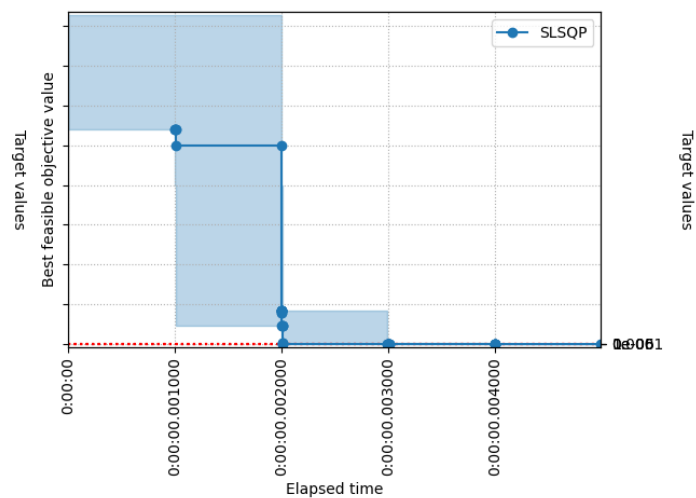
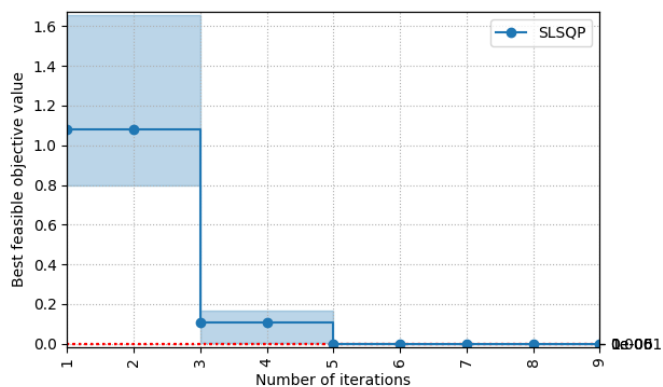
Rastrigin

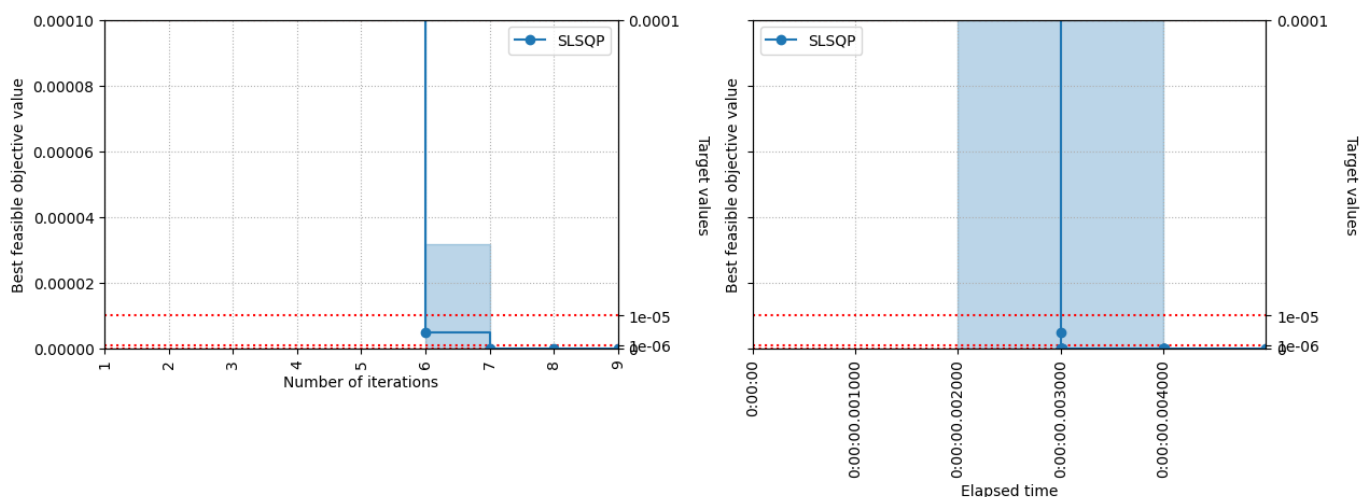
The results of the group of algorithms configurations 'SLSQP' for problem configuration 'Rastrigin'.

DATA PROFILES



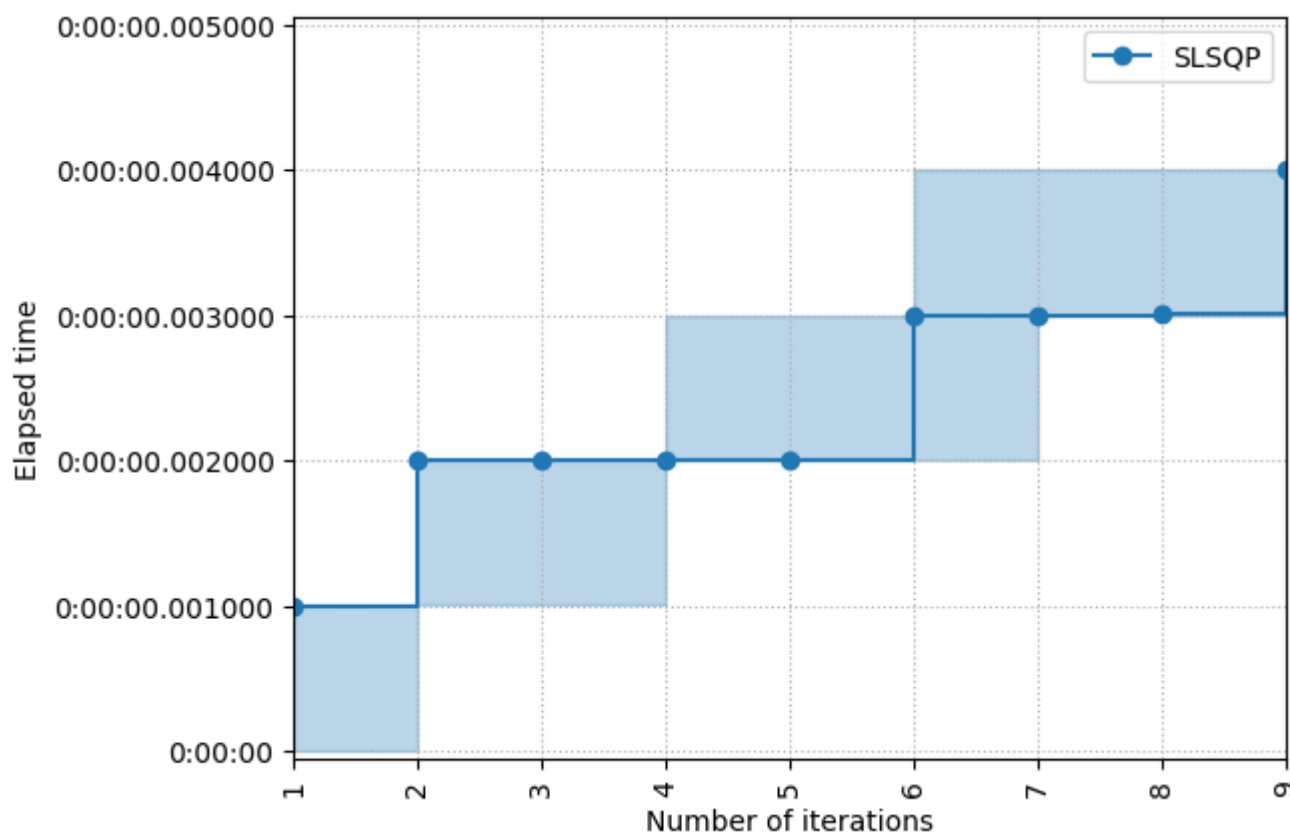
BEST FEASIBLE OBJECTIVE VALUE





	minimum	25th centile	median	75th centile	maximum
SLSQP	0	0	0	0	0

EXECUTION TIME



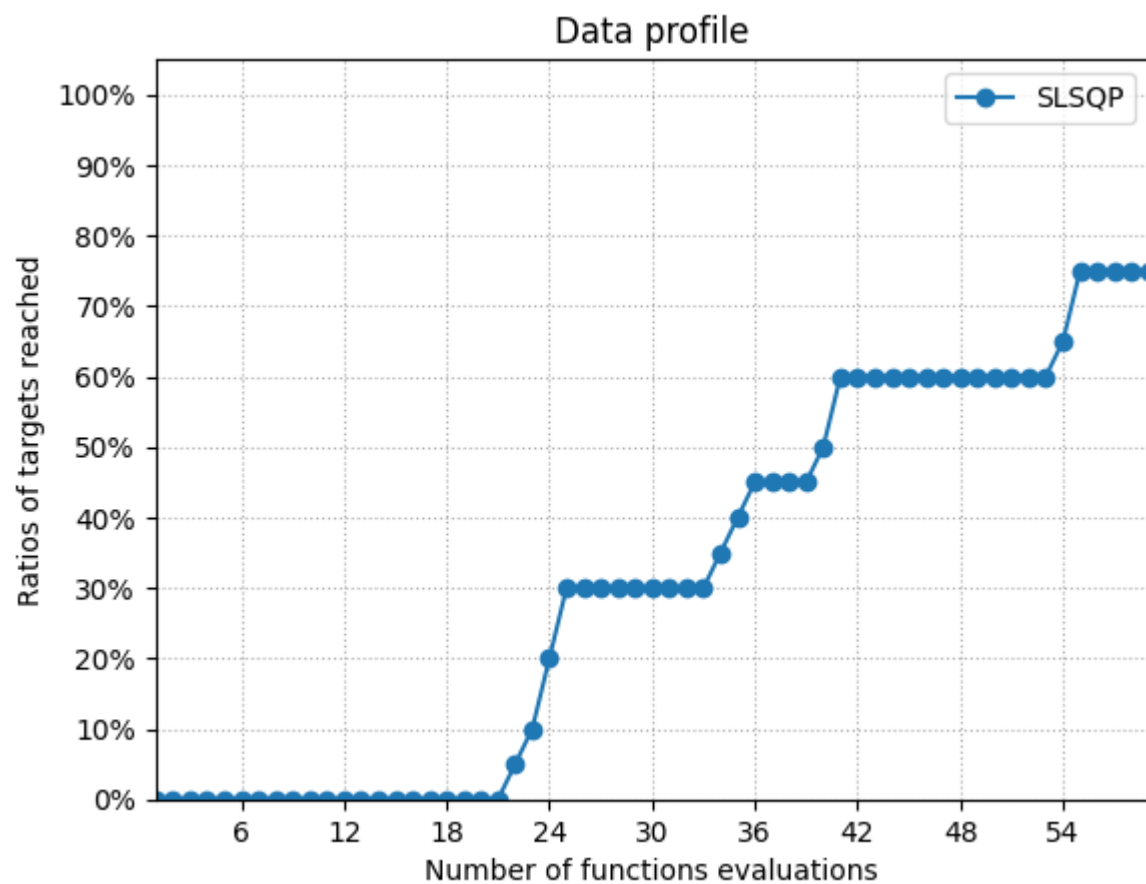
RESULTS FOR EACH ALGORITHM CONFIGURATION

- SLSQP

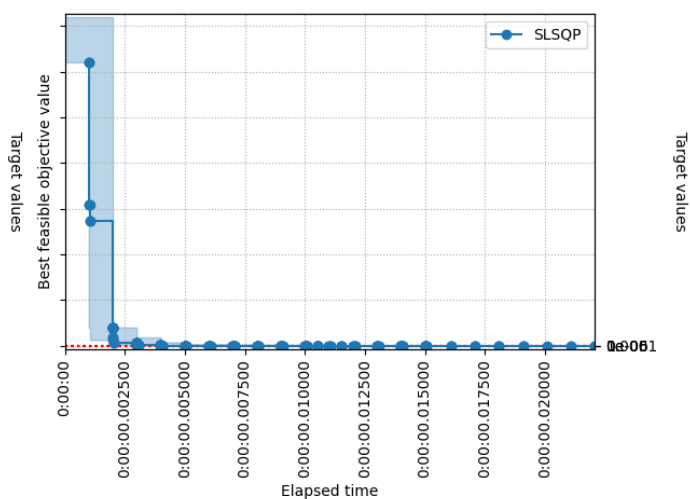
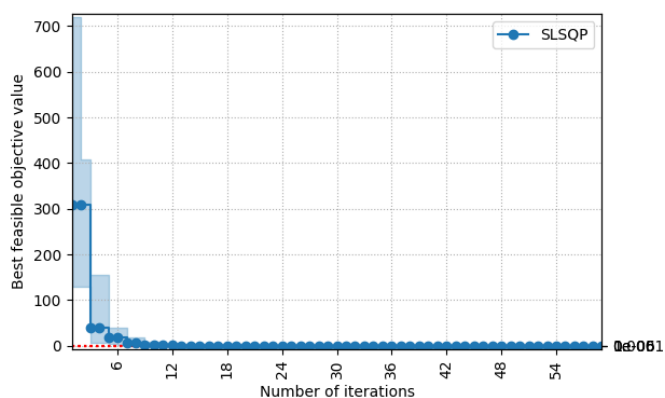
Rosenbrock

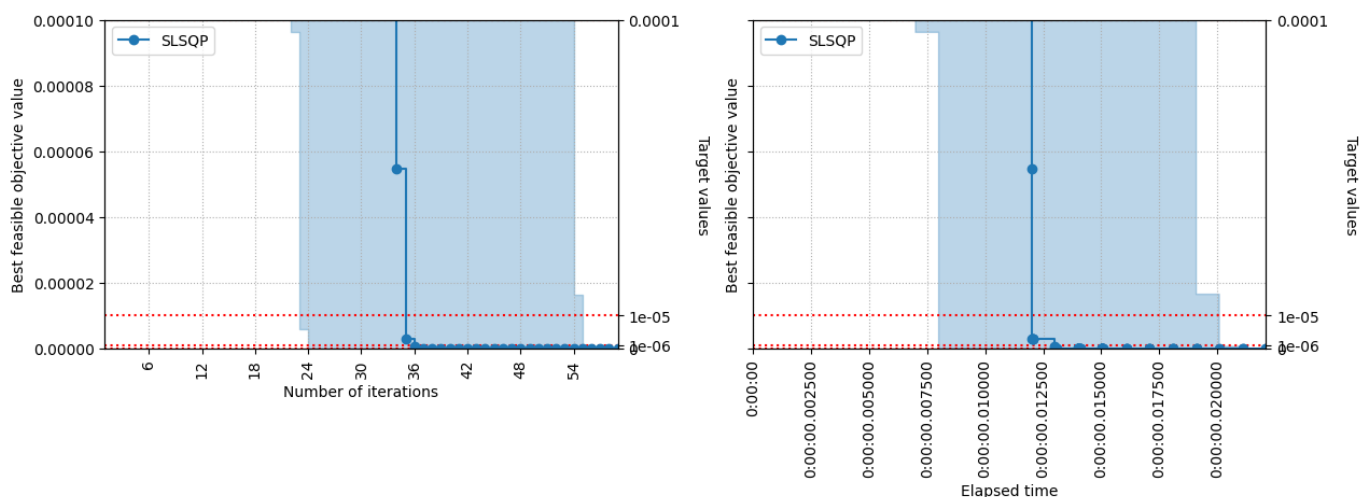
The results of the group of algorithms configurations 'SLSQP' for problem configuration 'Rosenbrock'.

DATA PROFILES



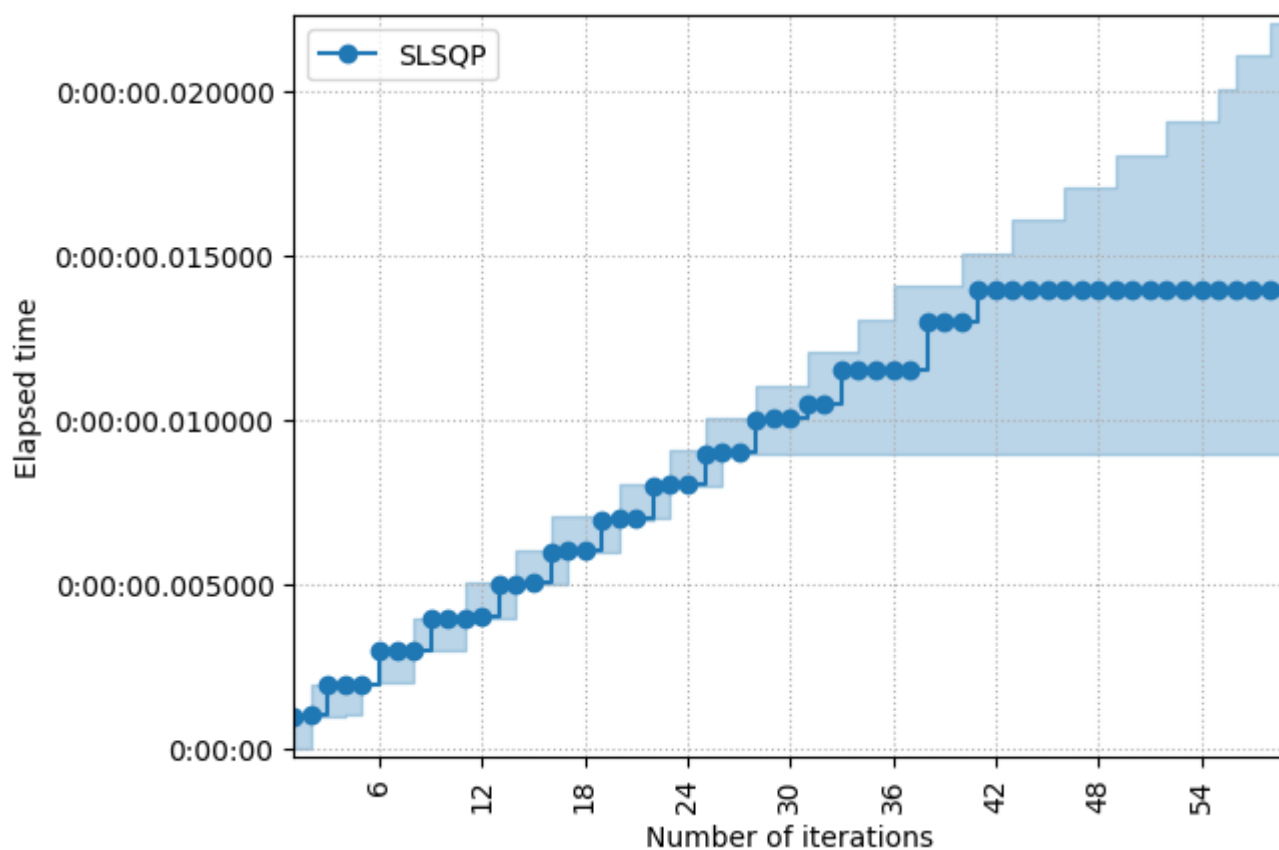
BEST FEASIBLE OBJECTIVE VALUE





	minimum	25th centile	median	75th centile	maximum
SLSQP	1.1824e-20	4.23227e-20	2.01609e-17	3.74191e-17	5.45617e-14

EXECUTION TIME



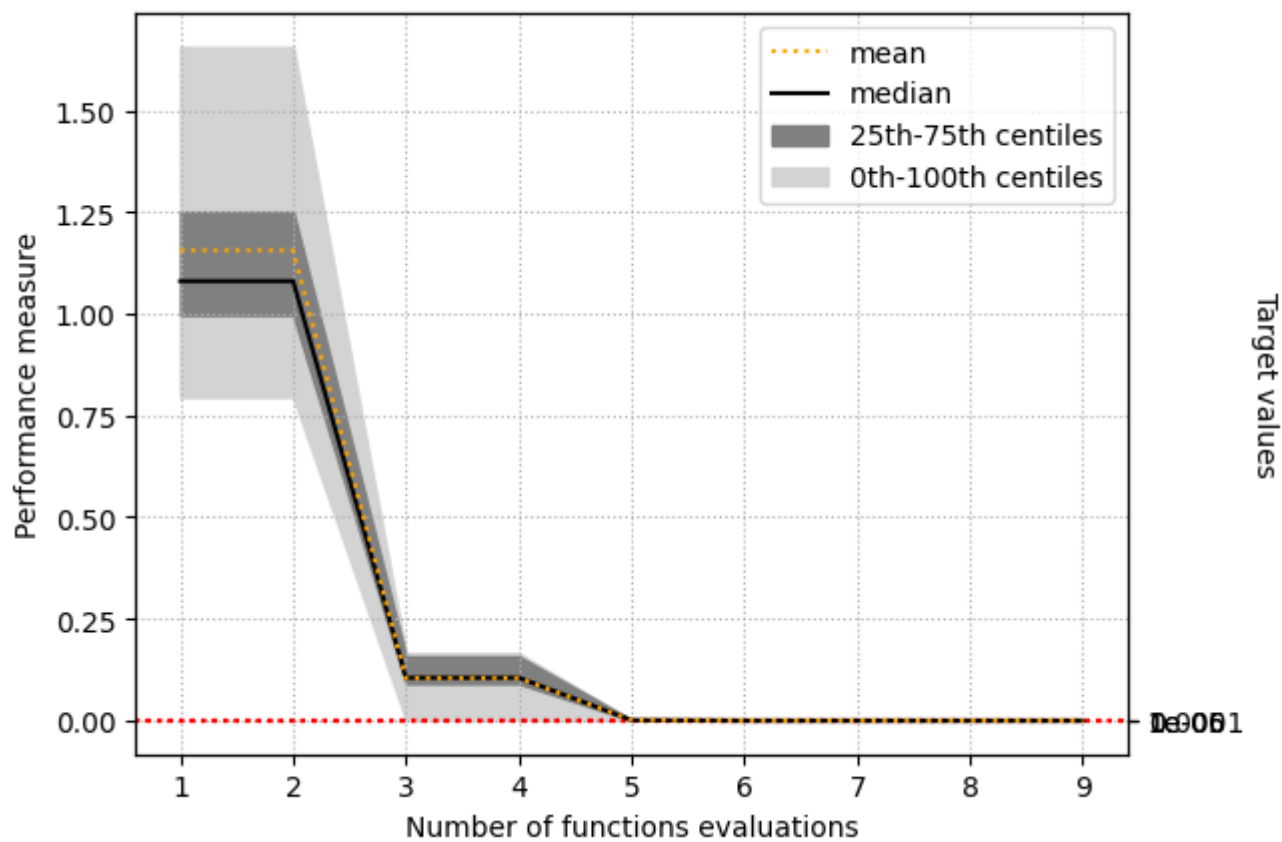
RESULTS FOR EACH ALGORITHM CONFIGURATION

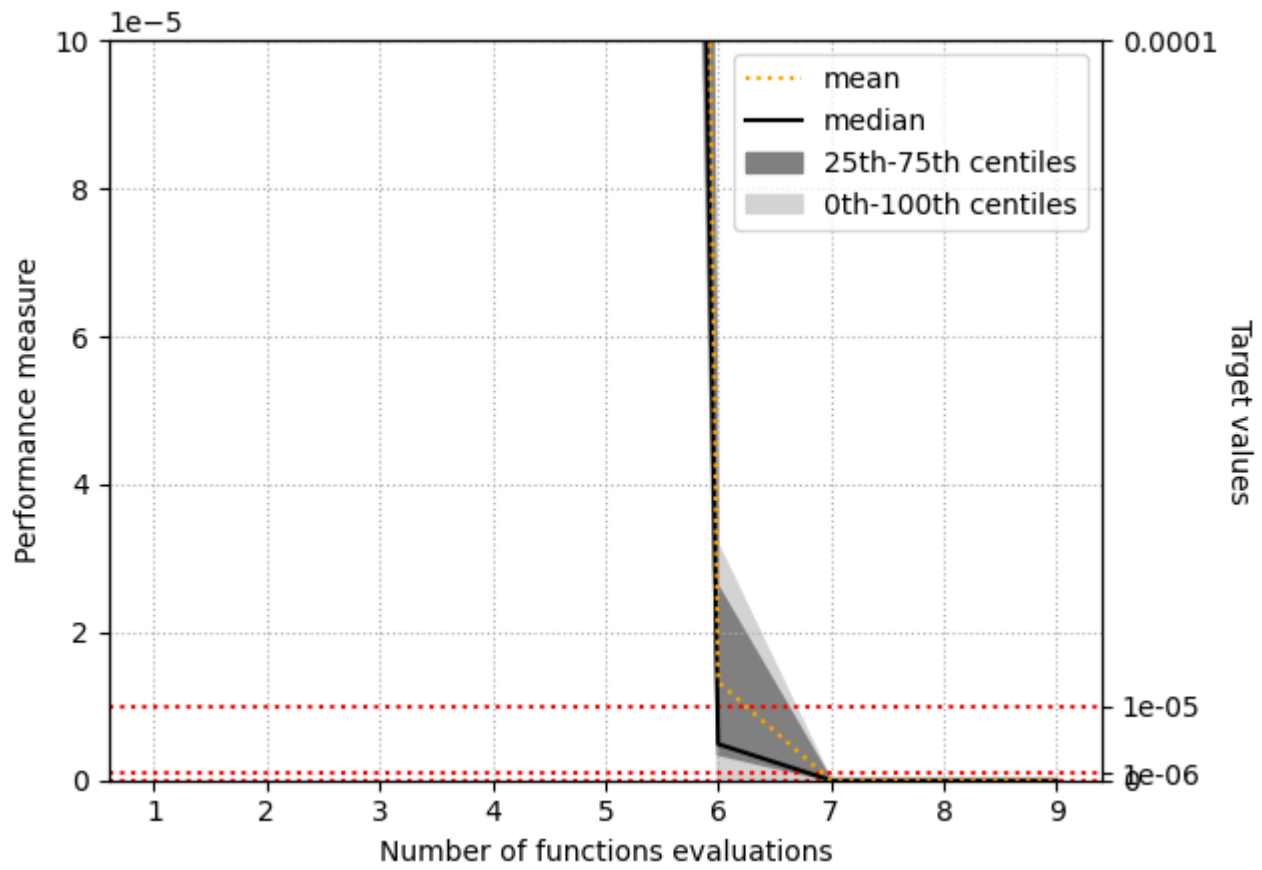
- SLSQP

Rastrigin

SLSQP ON RASTRIGIN

Performance measure



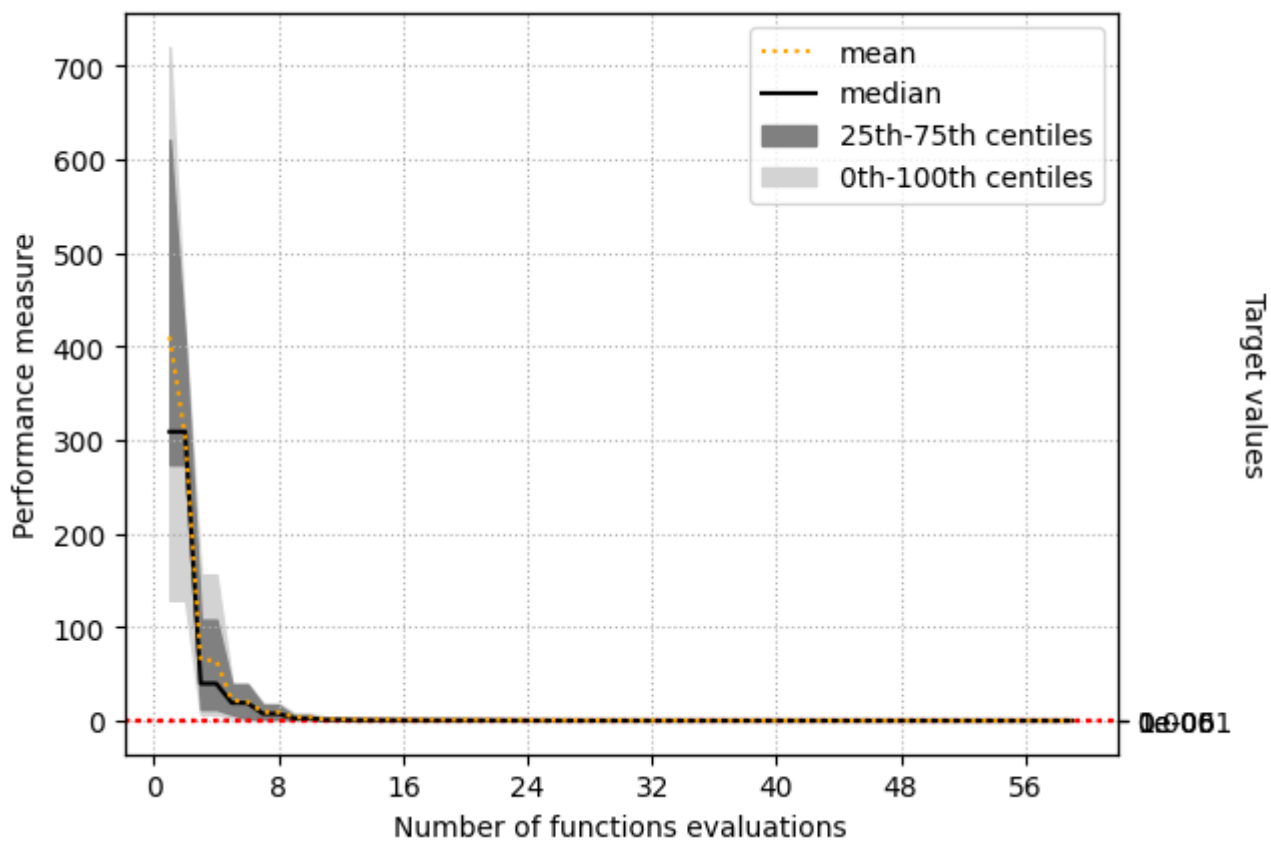


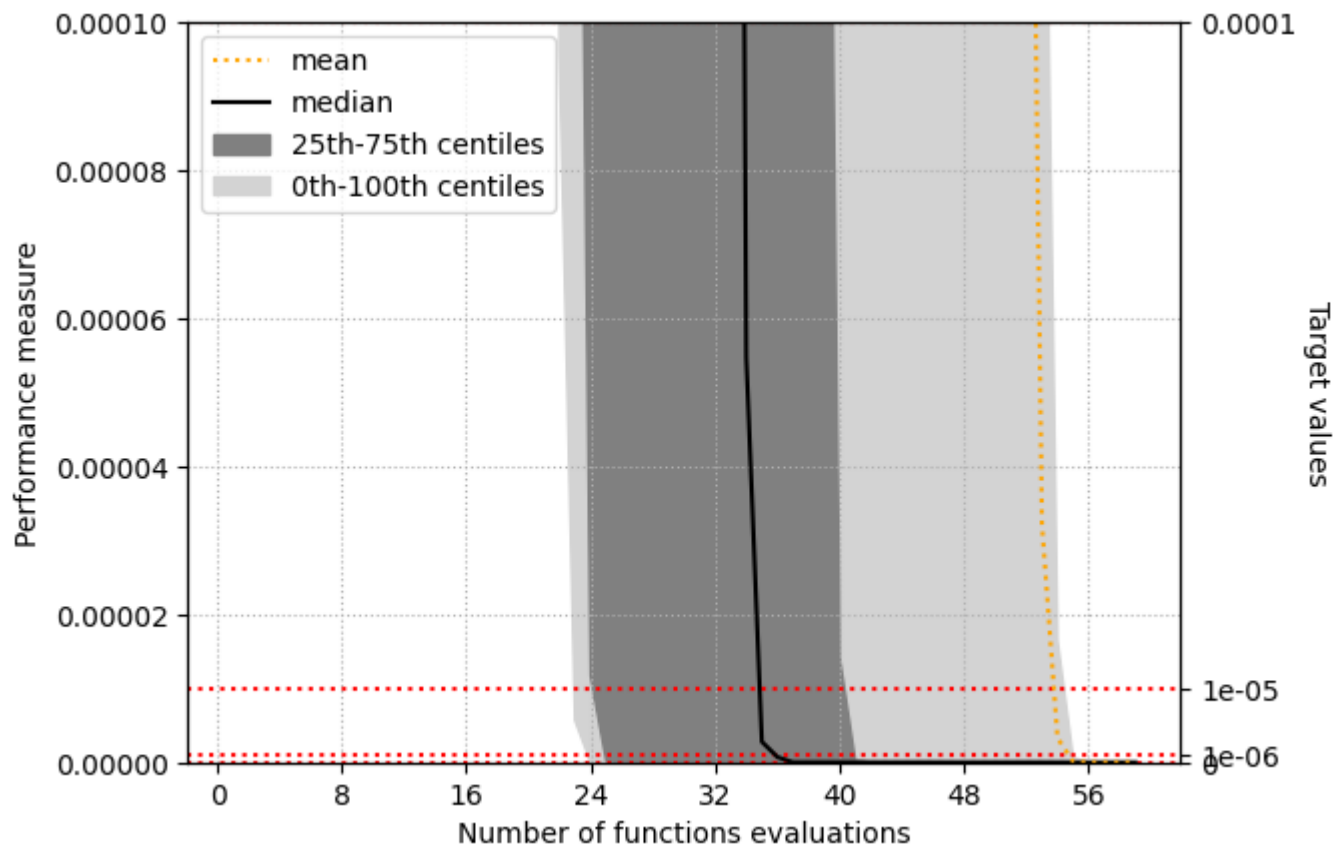
	SLSQP
maximum	0
75th centile	0
median	0
25th centile	0
minimum	0

Rosenbrock

SLSQP ON ROSENBRICK

Performance measure





SLSQP	
maximum	5.45617e-14
75th centile	3.74191e-17
median	2.01609e-17
25th centile	4.23227e-20
minimum	1.1824e-20